

N

NATIVE Describing a plant or animal indigenous to a particular area. See also *endemic*.

NATIVE ELEMENT A chemical element that is found in its pure state in nature. Copper, gold, sulfur, and tin can all be found in their native form. See also *mineral*.

NEAP TIDE The tide with the least difference between low and high water. Neap tides occur twice a month when the gravitational pull of the Moon and Sun on Earth's oceans are working against each other. See also *spring tide*.

NEEDLE-LEAF A needle-shaped leaf typical of coniferous trees. See also *coniferous*.

NOCTILUCENT CLOUD The highest clouds in Earth's atmosphere, illuminated at night when the Sun is below the horizon and lower levels of the atmosphere are in shade. These ice clouds are usually at an altitude of 47 to 53 miles (76 to 85 km). See also *mesosphere*.

NORMAL FAULT An angled fault where one side (the hanging wall) has slipped down relative to the other side (the footwall). See also *fault* (panel).

NUNATAK A mountain peak rising out of an icefield or ice sheet. See *icefield*, *ice sheet*.

O

OASIS An isolated fertile area within a desert, usually fed by water from a spring. See also *aquifer*.

OCCCLUSION In meteorology, the situation where a cold front overtakes a warm front, pushing a mass of warm air up and away from Earth's surface. There are two types: cold occlusions and warm occlusions. See also *front*.

OCEAN TRENCH A long, deep section of the ocean floor, usually lying above a subduction zone. See also *deep-sea trench*, *subduction*.

OCEANIC CRUST The outermost layer of Earth's lithosphere beneath the oceans. It is formed at spreading centers at oceanic ridges and is mainly made of basalt. See also *continental crust*, *crust*, *seafloor spreading*, *spreading ridge*.

OFFSHORE BAR A submerged or partly exposed ridge of sand or gravel away from and roughly parallel to the coast. Offshore bars are built up by waves and currents. See also *bar*.

OOZE The fine sediment covering a huge part of the deep ocean floor. Much of it is made up of the skeletal remains of microscopic marine life forms, such as foraminiferans and radiolarians.

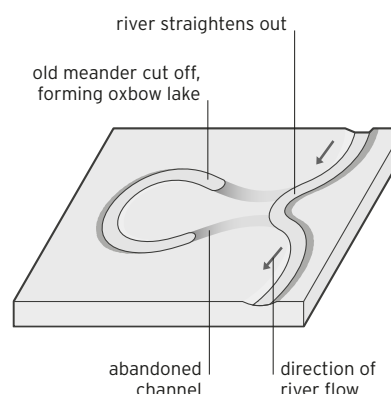
ORE A rock from which a metal or valuable mineral can be profitably mined.

OROGENESIS The process of mountain-building that results from massive horizontal pressures as tectonic plates push together.

OROGENY A mountain-building event resulting from the collision of tectonic plates, when sections of Earth's crust are compressed, folded, and faulted. See also *tectonic plate*.

OUTLET GLACIER A mass of slow-moving ice that flows from an ice sheet, ice cap, or icefield before following a valley. See also *valley glacier*.

OXBOW LAKE ▼ A body of water that forms when a meander is completely cut off from the main channel of a river. See also *meander*.



OVERTURNED FOLD A fold in which both limbs dip in the same direction. Overturned folds occur in areas of intense deformation of the rock layers. See also *fold* (panel).

P

PACK ICE A mass of ice floating on the sea, formed from many smaller pieces that have frozen together. See also *sea-ice*.

PAHOEHOE LAVA A type of basalt lava with a smooth or gently undulating surface. It often changes to a'a lava as it flows downslope. See also *a'a lava*, *block lava*.

PANCAKE ICE Small, flat, roughly circular areas of floating ice with curled margins where the "pancakes" have bumped into each other. Pancake ice is an early stage in the development of sea-ice. See also *sea-ice*.

PARABOLIC DUNE A U-shaped or V-shaped mound of fine to medium sand with long "arms" trailing upwind. Unlike barchans,

the steepest side (slip face) is on the convex side. See also *barchan*, *dune* (panel).

PARTIAL MELTING The process by which some minerals (in a rock being heated to great temperatures) melt, while others remain solid. This happens because some minerals have a lower melting point than others. See also *igneous rock*.

PEAT Partly decomposed plant material that accumulates in bogs, fens, and mires. Peat has a low mineral content.

PENINSULA An area of land protruding into a lake or sea.

PERIGLACIATION The processes that take place as a result of the regular freeze-thaw cycle in high-latitude or high-altitude regions where there is no permanent ice cover. Features include ice wedges and pingos. See also *ice wedge*, *permafrost*, *pingo*, *tundra*.

PERMAFROST Rock or soil that remains frozen for at least two years. Permafrost is characteristic of periglacial and glacial regions. See also *periglaciation*, *tundra*.

PHREATIC ERUPTION An explosive eruption that occurs when magma or hot rock come into contact with groundwater or surface water, turning it to steam.

PHYTOPLANKTON The tiny photosynthesizing life forms that float in the upper, sunlit layer of a lake or ocean and form the foundation of most aquatic food chains. See also *zooplankton*.

PIEDMONT GLACIER An area of slowly flowing ice that forms from the convergence of a number of valley glaciers at the foot of a mountain range. See also *valley glacier*.